

Group Problem Set 27: Rotational + Translational Motion

1. What are the rotational inertias of an empty toilet paper roll ($R = 0.025\text{m}$, $L = 0.1\text{m}$, $m = 0.10\text{kg}$) and a full toilet paper roll ($R_1 = 0.1\text{m}$, $R_2 = 0.025\text{m}$, $L = .1\text{m}$, $m = 1\text{kg}$). The toilet paper roll is rolled down a ramp 1 m off the ground.
 - (a) Write the general conservation of energy equations for each item. Be sure to take into account the mass distributions in the equations for inertia.
 - (b) What is the difference in start height required for the toilet paper rolls to have the same velocity at the bottom of the ramp?
 - (c) What is the final velocity at the base of the ramp of each TP if they start from a height of 1 m?
 - (d) How far will each roll travel if they both continue to roll on a flat surface with a coefficient of friction of 0.1.
 - (e) Write the conservation of energy equations for each roll as it rolls across the surface.
 - (f) Looking only at the conservation of energy equations, which portion of each roll's trajectory is conserved?
 - (g) Why?

2. A ball of radius 0.1m and mass of 1 kg, starts from a height of 1 m, and rolls down a hill.
- (a) What is the velocity of the ball at the bottom of the ramp.
 - (b) The ball hits a stationary ball at the bottom of the ramp. The stationary ball has a mass of 2 kg and radius of 0.05 m. What is the final velocity of the second ball after impact if the first ball stops and the collision is completely elastic?
 - (c) What is the final height of the second ball if it rolls up a ramp after the collision?
 - (d) What would the mass need to be, if instead, the first ball hit a second ball with the same radius (0.1 m) and stopped as the second ball continued up the ramp to a final height of 2 m?

3. Two children (25 kg each, assume equal mass distribution of a bar, with length 1m) hold hands and spin in a circle. They have an angular velocity of 1 rad/s.
- (a) What will happen to their angular velocity if they pull each other into a tighter formation, reducing their effective length to 0.6 m?
 - (b) What is their angular velocity?
 - (c) What is their initial total kinetic energy?
 - (d) What is their final total kinetic energy?
 - (e) Is their motion conserved?